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Radar Signal Deinterleaving in Electronic Warfare Systems: A Combined Approach

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ABSTRACT Traditional deinterleaving algorithms first cluster the pulses in frequency, pulse width, and angle of arrival parameter space. Then, they utilize time of arrival analysis techniques. This results in a performance loss, since these processes are not jointly combined; rather, they are sequentially connected. Additionally, they perform poorly in the presence of radars with staggered pulse repetition interval type due to abrupt time difference variations. To alleviate the suboptimality in the design and enhance the detection performance in the staggered type, we propose a combined approach based on the pulse repetition interval transform technique. It does not cluster the pulses as the traditional methods, instead, it combines all pulses by distance weighting. It deals with the detection problem in the staggered type by clustering the phases of the autocorrelation function. We develop an optimal weighting mechanism to fuse the outputs from phase clusters. Numerical results show that our method outperforms the traditional methods in different detection metrics via multiple scenarios, which is also validated by the theoretical findings.

INDEX TERMS Blind source separation, Clustering algorithms, Computational efficiency, Detection algorithms, Electronic countermeasures, Electronic warfare, Pattern clustering, Pattern recognition, Radar, Radar detection, Statistical analysis

I. INTRODUCTION

ELECTRONIC support measure (ESM) and intelligence (ELINT) systems operate on the received radar signals without transmission [1]. First, pulses of the received signal are detected and then transformed into pulse description words (PDW) [2]. PDWs include parameters such as time of arrival (TOA), frequency (RF), pulse width (PW), and angle of arrival (AOA). If the signals are sufficiently separated from each other in the RF-PW-AOA parameters domain, a simple clustering algorithm can be successful without the TOA analysis [3–10]. However, radars can have intersecting parameter ranges, and some of the parameters might be missing in the PDWs. Therefore, TOA analysis is a must for the deinterleaving algorithm [11].

SDIF is a well-known TOA analysis approach that utilizes the sequential differences of the TOAs (DTOA) [12]. The method iteratively takes the first, second, and so on differences and builds test statistics for each. Then, it thresholds the built test statistic to form the deinterleaving hypothesis. It works well in simple cases, but its performance degrades drastically as the number of existing radars increases or the pulse repetition interval (PRI) varies. PRI transform (PRIT-I) is an-

other well-known approach. It builds its test statistic by using the complex autocorrelation function of the pairwise DTOA values [13]. It performs poorly on varying PRI patterns, even though it detects the constant PRI type perfectly without any harmonics. To overcome this problem, an improvement in the method was proposed (PRIT-II) by utilizing the time origin shifts and overlapped histogram bins [14]. This advancement comes at a price, as we have shown separately it is likely to produce harmonics at high PRI levels [15].

Radars determine their PRI types according to their goals. Standard search and track radars prefer constant PRI, which aims to improve the Doppler resolution [16]. However, since the pulses are uniformly spaced, the constant PRI raises range-Doppler ambiguities and is considered to possess low electronic counter-countermeasures (ECCM) capabilities [17], [18]. Radars use random PRI types to overcome these issues, i.e., jittered PRI. Moreover, moving target indicator (MTI) and synthetic aperture (SAR) radars utilize the staggered PRI, a repeated PRI pattern, to expand the unambiguous Doppler interval [19–21]. Lastly, identification friend or foe (IFF) and beacon interrogation systems may transmit the pulses in bursts called pulse groups to improve range/velocity

resolution [22]. For PRI transform, constant, pulse group and jittered PRI types result in complex autocorrelation functions consisting of exponentials with similar phases. Summing exponentials with similar phases yields high test statistics to achieve their detections. However, the staggered PRI type causes the phases of the exponentials to toggle and become dissimilar. PRIT-I and PRIT-II generate much lower test statistics due to these phase transitions; therefore, missed detections occur [23], [24]. Different approaches, including the joint usage of SDIF and PRI transform, heuristic approaches using expert knowledge, and automata theory, were developed to enable the detection of the staggered PRI type. [23–25], [25–31].

Modern electronic warfare (EW) environment is highly complex such that radar signals with different PRI types are simultaneously intercepted by the ESM receiver. Due to the complex environment, some PDW parameters may be measured incorrectly by the receiver. The quality of the PDW parameter measurements also depends on the instantaneous bandwidth of the receiver and pulse-on-pulse situation [32]. Additionally, quantization and synchronization errors of the receiver can lead to high noise in the PDW [33]. Moreover, some radars (i.e. low probability of intercept radars) may prefer transmitting with low power to be invisible in the field or the receiver intercepts the radar pulses at its sensitivity level, which result in missing pulses for the deinterleaving algorithm [11], [28]. All these degradation examples have a great impact on the performance of the deinterleaving algorithm [6], [34] and our article's primary goal is to design a deinterleaving algorithm that can resolve highly complex EW environments.

Recently, deep learning solutions has taken much interest for ESM problems. One of the techniques to enable deep learning in the deinterleaving problem is to transform the interleaved radar signals into image data. In this way, time-series source separation problem is turned into an image processing problem. In [15], [35], [36], image segmentation models based on U-Net and Refine-Net were developed [37], [38]. A denoising autoencoder model, which is robust against missing pulse and PRI jitter, was developed [39]. Furthermore, some works provide multi-tasking solutions for the pulse detection, deinterleaving, recognition, and classification problems. In [40], a convolutional neural network-based method is proposed to solve the distortions emerged in the pulse extraction process by working on the intra-pulse measurements. The method utilizes the continuous wavelet transform for feature extraction and is capable of PRI deinterleaving and modulation recognition. [41] proposes a complete framework for pulse detection and modulation classification by utilizing time-frequency analysis and convolutional neural networks. [42] develops a recurrent neural network-based technique that focuses on joint radar signal classification and deinterleaving. All of these methods suffer from high complexity as they either work on intra-pulse measurements or they depend on big neural models to achieve their goals. Hence, they are not fit in real-time systems that are usually

expected to meet radar pulses up to a million per second.

Traditional deinterleaving algorithms first cluster the PDWs in the RF-PW-AOA parameters domain and then use TOA analysis [5], [40], [43–45]. This results in a performance loss, since the methods are not jointly combined rather they are sequentially connected [46]. If the noise is high or the radar has hopping RF/PW characteristics, pulses are incorrectly separated by the clustering operation and the performance loss gets amplified. To alleviate this suboptimality, enable the detection of the staggered PRI, and enhance the overall performance of the PRI transform, we propose Combined PRI Transform (CPRIT). Instead of RF-PW-AOA clustering, it weights the distance of the pulses in RF-PW-AOA parameter space by an inverse exponential function. To resolve the phase transitions disrupting the autocorrelation function in the staggered PRI, CPRIT clusters the phase values within the PRI transform algorithm. Each cluster has independent test statistics, so the destructive effect of summing exponentials with dissimilar phases does not happen. To fuse the test statistics coming from different clusters, we develop an optimal weighting function that minimizes the variance of the estimator. In this way, all clusters contribute to the final test statistic, which is compared to the threshold at the end. We explain the mathematical properties of CPRIT and demonstrate the role CPRIT plays between PRIT and SDIF depending on the distance parameter of the clustering algorithm. We also share results with simulation data, where CPRIT is compared to SDIF and PRIT via multiple scenarios including different PRI-RF-PW-AOA types. We show that the proposed approach enables the detection of the staggered PRI and almost always reaches the highest correct detection rates while yielding low false alarm rates.

Our contributions to the literature are the following.

- 1- CPRIT provides a novel deinterleaving approach that enables joint usage of clustering and PRI analysis methods. In this way, it alleviates the suboptimality observed in the traditional methods, which emerges from the sequential usage of these methods.
- 2- CPRIT develops an inverse exponential weighting function to combine the pulses. Different from the traditional methods that separate the pulses in the clustering section, it combines all pulses with their calculated weights.
- 3- CPRIT develops a unique test statistic function by applying an optimal weighting scheme that minimizes the variance of the estimator. Hence, phase clusters are optimally fused before the detection process.
- 4- CPRIT has several interesting properties, and we mathematically demonstrate them in the paper. Test statistic generated by the CPRIT is theoretically analyzed, which validates the numerical results.
- 5- CPRIT is numerically shown to enhance the overall detection performance of PRIT. It enables the detection of the staggered PRI and outperforms traditional deinterleaving algorithms in terms of different detection metrics.

The paper is organized as follows: Section II explains the

preliminary information. Section III describes the proposed method in detail. Section IV provides the theoretical analysis, Section V demonstrates the simulation results, and the last section is spared for conclusions. Throughout the paper, vectors, and matrices are denoted as $\mathbf{v}$, $\mathbf{V}$, while their elements are shown by v_i and $V_{i,j}$, respectively.

II. PRELIMINARY

A. PRI TYPES

Most of the radars follow similar PRI patterns to achieve their objectives. These patterns can be put into five classes: constant, staggered, pulse group constant (PGC), pulse group jittered (PGJ), and jittered. Constant PRI means using the same PRI, staggered and PGC PRI are the periodic PRI patterns. Compared to staggered PRI, pulses are transmitted as bursts and long silences in the case of PGC. Like PGC, pulses are also transmitted in bursts, in the case of PGJ. However, one PRI value in the sequence, usually the largest PRI, is not constant. Lastly, the jittered PRI type corresponds to highly varying PRI schemes. Figure 1 demonstrates the PRI patterns when no missing or erroneously detected pulse exists.

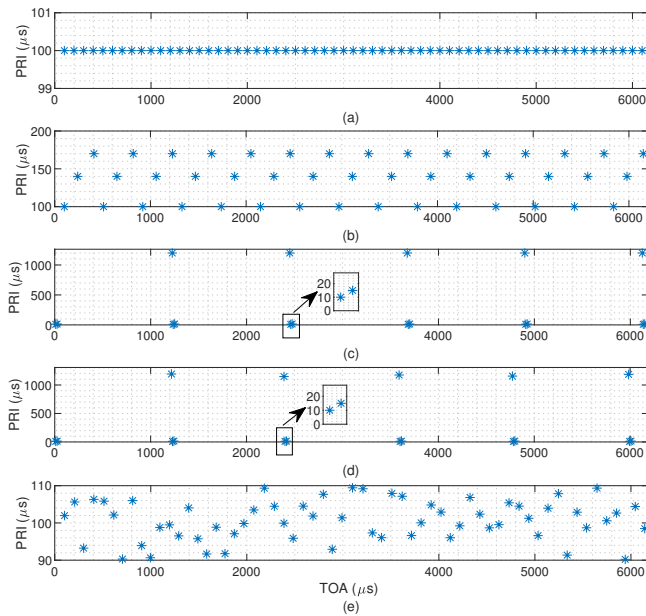


Figure 1. PRI types: (a) constant, (b) staggered, (c) PGC, (d) PGJ, and (e) jittered PRI patterns are shown from top to bottom. PGC and PGJ patterns are zoomed in additional windows to demonstrate the low PRI values corresponding to the bursts.

B. RF-PW-AOA TYPES

RF-PW types differ from the PRI types. We cover three types, as shown in Figure 2. Similar to PRI patterns, constant RF-PW means using the same values, and jittered RF-PW type corresponds to randomly varying patterns. Hopping RF-PW type is the usage of certain levels, instead of constant or random levels. Hence, the differences in the levels are mostly larger than a cluster can possess. Constant and jittered RF-PW types do not cause problems in the clustering stage of the

traditional deinterleaving algorithms, since all pulses end up in the same cluster before TOA analysis. However, hopping RF-PW type results in a significant error that separates pulses incorrectly in the clustering stage due to large sequential differences in the values, which disrupts the TOA analysis performance. Furthermore, we consider only constant AOA type. It is the most encountered case in the field, as positions of radars do not change in the input duration of the deinterleaving algorithm (usually in the orders of milliseconds).

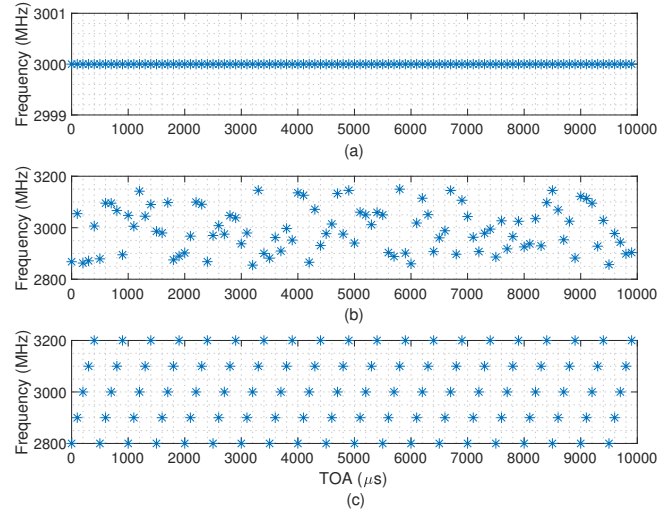


Figure 2. RF-PW types: (a) constant, (b) jittered, (c) hopping patterns are shown from top to bottom. We demonstrate frequency on the y-axis but the patterns are also valid for pulse width.

C. SDIF

SDIF is a TOA analysis method that utilizes the histogram of differences in TOAs [12]. If only one histogram value exceeds the threshold in the first difference, it is marked as detection. However, if many emitters are present, more than one histogram value will exceed the threshold. The histogram does not reveal the actual PRI value as they are hidden in higher differences. Then the next difference is calculated to reach the actual PRI values of the radars. Furthermore, subharmonic checking is conducted to avoid false alarms by limiting the upper histogram limit and extracting the first lowest value exceeding the threshold. Assuming that TOAs follow the Poisson distribution, the optimal threshold can be calculated as

$$\nu_{sdif} = q(N - c)e^{-\frac{\tau}{rB}}, \quad (1)$$

where N is the total number of pulses, K is the total number of histogram bins, c is the difference level, and τ is the PRI interval. q is the constant determining the maximum rate of the missing pulse, and r is the constant regarding the distribution of the TOAs. As [12] suggested, the parameters $q = 0.3$ and $r = 0.7$ are kept in the range $[0, 1]$ and experimentally determined in our simulations. Note that increasing their values yields a higher threshold value.

D. PRI TRANSFORM

The PRI Transform (PRIT) was initially designed to capture constant PRI patterns by utilizing the complex autocorrelation function [13]. The initial version (PRIT-I) was shown to estimate PRI of radars without generating harmonic PRI which are the integer multiples of the actual PRI. Harmonic PRI usually occur when the missing pulse ratio is high or the deinterleaving algorithm is designed to favor the detection against the false alarm. Moreover, PRIT-I is unable to detect varying PRI patterns, as the accumulative phase error increases in the complex autocorrelation. To resolve this issue, the second version PRIT-II, was developed [14]. These two methods will be summarized below.

1) PRI Transform-I [13]

We let N be the total pulse number. TOAs of the interleaved pulses are expressed as a sum of unit impulses,

$$g(t) = \sum_{n=0}^{N-1} \delta(t - t_n), \quad (2)$$

where t_n is the TOA of the n -th pulse, and $\delta(\cdot)$ is the Dirac delta function. PRI transform is calculated by the complex autocorrelation function of DTOA values as follows:

$$D(\tau) = \left| \sum_{n=1}^{N-1} \sum_{m=0}^{n-1} \delta(\tau - t_n + t_m) \exp \left[\frac{2\pi i t_n}{t_n - t_m} \right] \right|. \quad (3)$$

The phase factor $\exp(2\pi i t_n / (t_n - t_m))$ is the core idea behind the PRI transform, which generates high test statistics values from the interleaved radar pulses with the constant PRI type. The phase factor also suppresses the subharmonics, which would occur at the integer multiple of the actual PRI level. In Figure 3, we demonstrate the subharmonics by decomposing a pulse train with $\text{PRI} = \tau$. l subharmonics have the PRI level $l\tau$. Their start times come in sequential order, which causes their phase factors to vary. We let $\theta_n = 2\pi t_n / (t_n - t_m)$ and represent the phase factors $\exp(i\theta_n)$ of the subharmonic components by points on the unit circle as in Figure 4. It is seen that they are symmetrical around the origin, and the vector sum of the phase factors becomes zero, except $l = 1$. Therefore, if we multiply the right-hand side of (2) by the phase factor $\exp(2\pi i t_n / (t_n - t_m))$ and take the summation as in (3), the subharmonics in the autocorrelation function are suppressed.

The PRI transform defined by impulses as in (3) requires the summation of impulses, which is inappropriate to numerically calculate. Hence, we use the discrete version of the method that is computed in the range $[\tau_{\min}, \tau_{\max}]$, where $\tau_{\min}$ and $\tau_{\max}$ determines the bounds for the PRI levels that will be analyzed, respectively. Then, PRI bins are formed by having a constant bin width b with no overlap. The center of each PRI level is calculated as

$$\tau_k = (k - 1/2)b + \tau_{\min}, \quad k = 1, 2, \dots, K. \quad (4)$$

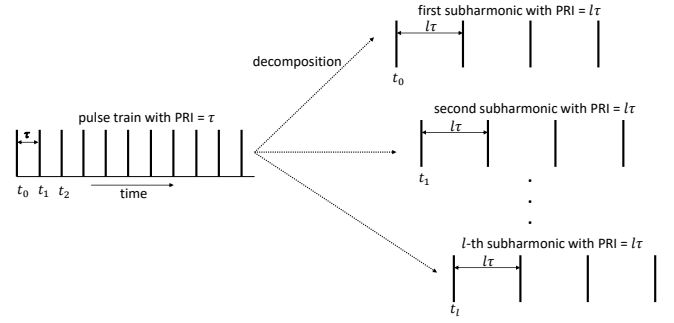


Figure 3. Subharmonic components of the original pulse train with PRI τ . Each has $l\tau$ PRI value, and their start times come in sequential order.

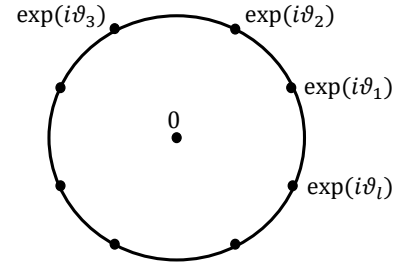


Figure 4. Phase factors of l subharmonic components. They are cyclic due to the complex exponential.

where K is the number of PRI bins or PRI levels. Lastly, the discrete PRI transform is calculated as:

$$D[k] = \left| \sum_{\substack{(m,n) \\ \{\tau_k - b_k/2 \leq t_n - t_m < \tau_k + b_k/2\}}} \exp \left[\frac{2\pi i t_n}{t_n - t_m} \right] \right|. \quad (5)$$

2) PRI Transform-II [14]

There are two main reasons for PRIT-I being unable to detect varying patterns. The first is the large accumulative phase error as the number of pulses increases. This reduces the peak level of the test statistic. The second is the peak power leak around the adjacent bins due to the usage of non-overlapping PRI bins. To overcome these two problems, the modified algorithm shifts the time origins for each pulse pair and utilizes overlapped PRI bins. The time origins are shifted as follows. First, the preliminary phase is determined by

$$\eta_0 = \frac{t_n - O_k}{t_n - t_m}, \quad (6)$$

where O_k is the previous time origin of the k -th PRI level. After that, the phase is decomposed as

$$\eta_0 = \nu(1 + \zeta), \quad (7)$$

where ν is an integer and ζ is a real number such that $-1/2 < \zeta < 1/2$. There are two conditions for the new origin assignment. If ν is equal to one and t_m equals to O_k , then t_n is assigned as the new origin. Additionally, if ν is larger than or equal to 2 and $|\zeta|$ is lower than or equal to ζ_0 , t_n is again assigned as the new time origin. Here, the positively valued parameter ζ_0 determines how mobile the time origins are.

Next, bins are overlapped with 50% to enhance the resolution and have a higher peak level. This is crucial for a correct PRI estimation. Centers of the PRI levels are calculated as

$$\tau_k = (k/2)b + \tau_{\min}, \quad k = 1, 2, \dots, K. \quad (8)$$

Finally, the test statistic $D[k]$ is compared to the following threshold

$$\nu_{\text{pfit}} = \max \left(\alpha \frac{T}{\tau_k}, \beta C_k, \gamma \sqrt{T \rho^2 b_k} \right). \quad (9)$$

PRIT is fed by a group of PDWs in each call. Therefore, it works on data that is collected for a certain duration. Here, T denotes the input duration. C_k is the valid pulse pairs that contribute to the test statistic of k -th PRI level. In other words, C_k shows the total number of pairs (m, n) that obeys $\tau_k - b_k/2 < t_n - t_m < \tau_k + b_k/2$. ρ is the pulse density and it can be calculated as N/T . Three constants are to be set. α is related to the required pulse density, β sets the threshold against subharmonics, and γ determines the tolerance against noise. In our simulations, we select $\alpha = 0.3$, $\beta = 0.15$, $\gamma = 3$, which are the suggested values in [14].

III. PROPOSED METHOD: COMBINED PRI TRANSFORM (CPRIT)

Traditional deinterleaving algorithms utilize RF-PW-AOA clustering, TOA analysis, and pulse tracking successively, as shown in Figure 5. First, pulses are clustered in RF-PW-AOA parameter space. Then, TOA analysis is conducted to estimate the PRI of the radar in each cluster independently. Lastly, estimated PRI is used to track and extract the radar pulses out of the interleaved sequence. This is a suboptimal option, since each cluster has different TOA analysis processes, and there is no connection between them. If the noise is high or the radar has hopping RF/PW characteristics, pulses are incorrectly separated by the clustering operation, causing missed detection in the TOA analysis section.

The proposed method does not accompany RF-PW-AOA clustering, as its flow is shown in Figure 6. Instead, it aims to get all pulse pairs to contribute to the test statistics. For this purpose, the pulse pair descriptor word (PPDW) generator first forms the pulse pair information including the distance weight, DTOA, and complex autocorrelation exponential $(\psi, DTOA, e^{i\theta})$. DTOA values which result from (t_n, t_m) pairs and obey $\tau_k - b_k/2 \leq t_n - t_m < \tau_k + b_k/2$ are grouped, where τ_k is the k -th PRI level. In each PRI level group, pulse pairs are clustered by the phase angles of the complex exponentials. Hence, pulse pairs having different phases, which is problematic in the staggered PRI case, are separated. Each cluster calculates its test statistic with the usage of the distance weights. Then, to reach the final test statistic and fuse clusters, an optimal weighting scheme is applied. So, target PRI values are estimated. Lastly, estimated PRI is used to track and extract the radar pulses out of the interleaved sequence. Thus, we combine all pulse pairs contributing to the test statistics by not using the RF-PW-AOA clustering, which removes the suboptimality in the design of

the deinterleaving algorithm. Overall, the proposed method brings three main novelties. The first is the removal of RF-PW-AOA clustering before the TOA analysis by using the distance weights. The second is the phase clustering to deal with the missed detection problem in the case of staggered PRI. The last is the optimal weighting function applied to fuse the test statistics from different phase clusters. We share the details below.

A. PULSE PAIR DESCRIPTOR WORD (PPDW) GENERATION

A PPDW consists of information regarding the relation between two pulses. Different from PDWs, which carry information about a single pulse, PPDWs indicate how two pulses are related and how close they are. A PPDW is formed by three parameters $(\psi, DTOA, e^{i\theta})$. First is the distance weight that is calculated by the absolute difference in PW, RF, and AOA parameters of two pulses. It is between 0 and 1, while large values indicate pulses being close. Specifically,

$$\psi = e^{-|\Delta_{rf}|/\epsilon_{rf}} e^{-|\Delta_{pw}|/\epsilon_{pw}} e^{-|\Delta_{aoa}|/\epsilon_{aoa}}. \quad (10)$$

Here, Δ_{pw} , Δ_{rf} , and Δ_{aoa} represent the difference in PW, RF, and AOA parameters of two pulses, respectively. ϵ_{pw} , ϵ_{rf} , and ϵ_{aoa} are the normalizing coefficients. We prefer $\epsilon_{pw} = 1 \mu s$, $\epsilon_{rf} = 10 \text{ MHz}$, and $\epsilon_{aoa} = 10 \text{ degrees}$ in our practice. We chose them a lot larger than the typical measurement errors of an ESM receiver since we aim to process radar signals having hopping RF/PW characteristics without separating them. The decaying exponential provides all pulse pairs to contribute to the final test statistic with different weights, which is contrary to traditional deinterleaving algorithms that separate pulses by the RF-PW-AOA clustering stage. Thus, weighting enables us to be resilient against noise and radar signals with hopping characteristics. The second parameter of a PPDW is DTOA, which is the difference in TOA parameters of two pulses $(t_n - t_m)$. This parameter is used in the PRI level grouping stage. Third is the complex autocorrelation $\exp[2\pi i t_n/(t_n - t_m)]$, which is used in the phase clustering stage.

B. PRI LEVEL GROUPING

PRI level grouping stage utilizes the second parameter of PPDWs, DTOA. There are $N(N-1)/2$ possible pulse pairs that are to be grouped. The goal of grouping is to analyze only the pulse pairs that generate similar DTOA values. Considering Figure 1, constant and jittered PRI values are grouped around their center values. However, staggered, PGC, and PGJ PRI groups exist at the period of the sequence. Grouping is achieved similar to the PRIT case. We form a matrix $\mathbf{E}$ of size $K \times N$, in which its rows consist of the third parameter of PPDWs, exponentials $\exp[2\pi i t_n/(t_n - t_m)]$, that correspond to a single PRI level group. Specifically, the k -th row is formed by the exponentials of the (t_n, t_m) pairs obeying $\tau_k - b_k/2 \leq t_n - t_m < \tau_k + b_k/2$, where τ_k is the k -th PRI level. Then, the phase angles of the exponentials in $\mathbf{E}$ are calculated and written in a new matrix $\mathbf{Q}$ of size $K \times N$.

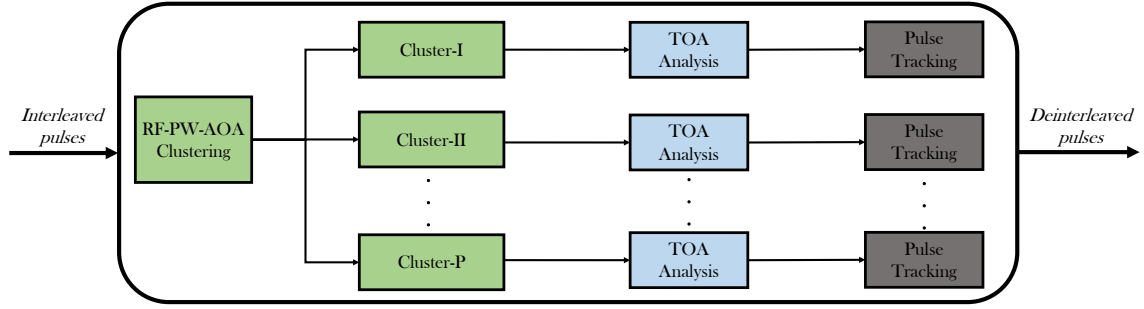


Figure 5. Traditional deinterleaving procedure. Pulses are first clustered in RF-PW-AOA parameter space. Then, TOA analysis and pulse tracking are conducted for each cluster.

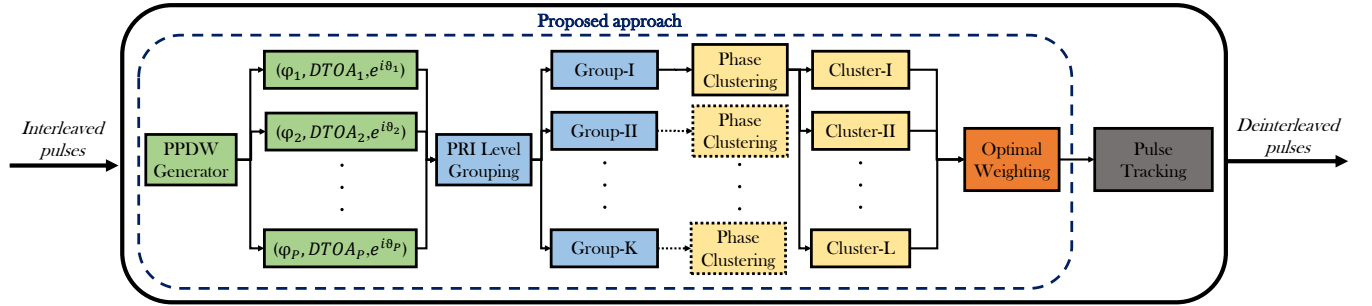


Figure 6. Proposed deinterleaving procedure. Pulse pair information is formed and PRI is grouped first. Then, phase clustering is applied for each PRI group. Optimal weighting fuses test statistics from different phase clusters. Lastly, pulse tracking is conducted.

The angle $Q_{k,j} \in [-\pi, +\pi]$ is defined as the following

$$Q_{k,j} = \text{atan2}(\text{Im}\{E_{k,j}\}, \text{Re}\{E_{k,j}\}), \quad (11)$$

where $\text{atan2}(\cdot)$ is the four-quadrant inverse tangent operation. After the PRI grouping, each group is then independently processed. In each group, phase angles of the complex autocorrelation are clustered.

C. PHASE CLUSTERING

In the case of the staggered PRI, phases of the exponentials in (5) mismatch and toggle. It leads to having a low test statistic $D[k]$ since it is the summation of these toggling exponentials. To prevent a reduction in the value of $D[k]$, which might cause missed detections, we propose to cluster the exponentials in each PRI group according to their phases and construct test statistics for each cluster individually. Hence, we obtain multiple test statistics for each PRI group, which differs from PRIT generating a single test statistic. The clustering operation necessitates a fusing technique to transform the multiple test statistics into one so that it can be compared to a certain threshold. We will discuss the details of the weighting mechanism in the next chapter.

Each row of $\mathbf{Q}$ is separately clustered. Note that dealing with the $-\pi \leftrightarrow \pi$ wrapping is necessary at this point. For the clustering algorithm, we prefer the Rough-DBSCAN algorithm [47], which is the faster version of the original DBSCAN [48]. Despite the noise and outliers, it can successfully discover any arbitrary shape of clusters as long as its parameters are specified correctly [49]. The most critical

parameter of the algorithm is the distance parameter. It indicates the maximum distance between two samples for each to be accepted as a neighbor of the other. Note that this is not a maximum bound on the range of points within a cluster. For our problem, the clustering distance parameter of the algorithm

$$\epsilon_Q = \frac{2\pi}{\tau^*}, \quad (12)$$

is set by the DTOA range $\tau^* = 1 \mu s$. The parameter is intentionally selected to be a lot higher than the value ($\approx 0.01 \mu s$) in which two TOA values are indistinguishable by the deinterleaving algorithm. In this way, we guarantee that similar DTOA values end up in the same cluster. The parameter is utilized by calculating the Euclidean distance. We cluster one-dimensional data, and Euclidean distance is known to perform well in this case, as there is no scaling issue. Another parameter of the algorithm is the minimum number of samples (or total weight) in a neighborhood for a point to be considered a core point. We set this parameter as three.

Up to this point, we have clustered the phase angles of the exponentials for each PRI level but have yet to generate the test statistic. Therefore, we calculate the test statistic by taking the absolute of the weighted summation of the exponentials belonging to the l -th cluster of the k -th PRI level as follows

$$G'[k, l] = \left| \sum_j \psi_{k,j} E_{k,j} \right|, \quad (13)$$

$$\{Q_{k,j} \in S_l\}$$

where S_l indicates the l -th cluster of the exponentials, and $\psi_{k,j}$ corresponds to the distance weight for the pulse pair. At this point, we have multiple test statistics for each PRI level, which is different from PRIT having a single test statistic as described in (5).

D. OPTIMAL WEIGHTING

Since our approach generates multiple test statistics from different phase clusters, a method to fuse them is required to reach a single test statistic that can be compared to the threshold. Therefore, we utilize a weighting procedure to obtain an optimal solution. Let the weighting coefficient be $w_{k,l}$ for the l -th cluster S_l of k -th PRI level τ_k . Then, the weighted test statistic becomes

$$G[k] = \sum_{l=1}^L w_{k,l} G'[k, l], \quad (14)$$

where L indicates the number of clusters. With the assumption of clusters being statistically independent, the variance of $G[k]$ can be calculated as

$$\text{var}(G[k]) = \sum_{l=1}^L w_{k,l}^2 \sigma_{k,l}^2. \quad (15)$$

Here, $\sigma_{k,l}^2$ is the variance of the test statistic corresponding to l -th cluster of the k -th PRI level. We minimize $\text{var}(G[k])$ under the constraint of $\sum_{l=1}^L w_{k,l} = L$, which leads to the following optimization problem

$$\min_{w_{k,l}, \lambda} \left[\sum_{l=1}^L w_{k,l}^2 \sigma_{k,l}^2 - \lambda \left(\sum_{l=1}^L w_{k,l} - L \right) \right], \quad (16)$$

where λ is the Lagrange multiplier. By equating the gradient with respect to the parameters of (16) to zero, we reach the following

$$\lambda = \frac{2L}{\sum_{l=1}^L \sigma_{k,l}^2}, \quad (17)$$

$$w_{k,l} = \frac{L}{\sigma_{k,l}^2 \left(\sum_{l=1}^L \frac{1}{\sigma_{k,l}^2} \right)}. \quad (18)$$

Substituting (18) into (14) provides the final test statistic of the proposed method

$$G[k] = \sum_{l=1}^L \frac{L}{\sigma_{k,l}^2 \left(\sum_{l=1}^L \frac{1}{\sigma_{k,l}^2} \right)} G'[k, l]. \quad (19)$$

Hence, we reach the inverse-variance weighting function (IVW), which minimizes the variance of the estimator and equivalently maximizes the likelihood function [50]. Note that IVW puts more emphasis on the clusters composed of exponentials with similar phases having low variance. Additionally, it is easy to see that $G[k] = \sum_{l=1}^L G'[k, l]$ if $\sigma_{k,1}^2 = \sigma_{k,2}^2 = \dots = \sigma_{k,L}^2$. In the end, $G[k]$ is compared to the threshold defined by (9).

In summary, we develop distance weighting that combines all pulse pairs to replace RF-PW-AOA clustering in the tra-

Algorithm 1 Combined PRI Transform (CPRIT)

Input: τ : PRI levels $[\tau_1, \tau_2, \dots, \tau_K]^T$, $\mathbf{t}$: TOAs $[t_1, t_2, \dots, t_N]^T$, $\mathbf{rf}$: RFs $[rf_1, rf_2, \dots, rf_N]^T$, $\mathbf{pw}$: PWs $[pw_1, pw_2, \dots, pw_N]^T$, $\mathbf{aoa}$: AOAs $[aoa_1, aoa_2, \dots, aoa_N]^T$, $\epsilon_{pw}, \epsilon_{rf}, \epsilon_{aoa}, \epsilon_Q$

- 1: Initialize $\mathbf{C} = \mathbf{0} \in \mathbb{R}^{K \times 1}$ $\triangleright$ counter for the number of valid pulse pairs per each PRI level
- 2: Initialize $\mathbf{E} = \mathbf{0} \in \mathbb{R}^{K \times N}$
- 3: **for all** (n,m) **do**
- 4: **for all** k satisfying $\tau_k - b_k/2 \leq t_n - t_m < \tau_k + b_k/2$ **do** $\triangleright$ for each PRI level
- 5: $C_k = C_k + 1$
- 6: $\Delta_{rf} = rf_n - rf_m$
- 7: $\Delta_{pw} = pw_n - pw_m$
- 8: $\Delta_{aoa} = aoa_n - aoa_m$
- 9: $\psi_{k,c_k} = e^{-|\Delta_{rf}|/\epsilon_{rf}} e^{-|\Delta_{pw}|/\epsilon_{pw}} e^{-|\Delta_{aoa}|/\epsilon_{aoa}}$
- 10: $E_{k,c_k} = \exp[2\pi i t_n / (t_n - t_m)]$
- 11: $Q_{k,c_k} = \text{atan2}(\text{Im}\{E_{k,c_k}\}, \text{Re}\{E_{k,c_k}\})$
- 12: **end for**
- 13: **end for**
- 14: **for** $k = 1 : K$ **do** $\triangleright$ for each PRI level
- 15: Cluster the angles in $Q_{k,:}$ using Rough-DBSCAN with parameter ϵ_Q .
- 16: **for all** l **do** $\triangleright$ for each cluster
- 17: Calculate $G'[k, l]$ according to (13).
- 18: **end for**
- 19: Calculate $G[k]$ according to (19).
- 20: **end for**
- 21: Compare $G[k]$ s to the threshold given by (9). Mark the PRI levels exceeding the threshold.

ditional deinterleaving algorithms. Additionally, we utilize phase clustering to obtain higher test statistics in the staggered PRI case and apply an optimal weighting procedure to fuse the test statistics from different phase clusters. Algorithm 1 demonstrates the pseudocode of CPRIT.

IV. THEORETICAL ANALYSIS

In this section, we aim to make a theoretical analysis to investigate how the proposed technique affects the test statistic value of PRIT. We first share the statistical properties of the test statistics of CPRIT, in which we calculate the mean and variance of the test statistic. Then, we make propositions that reveal some properties of the CPRIT with two separate modes. Both modes follow the same procedure in Figure 6 with a slight difference. The first one does not employ the optimal weighting (equal weighting is used to fuse the phase clusters), while the second does. Hence, we demonstrate the effects of phase clustering and optimal weighting individually. We deduce that they have different roles in generating the test statistic. Specifically, the phase clustering enables CPRIT to have higher test statistic values, and the optimal weighting causes the variance of the test statistic to be minimized, which agrees with our simulation results that we will share in the next section. Additionally, we prove the role CPRIT plays

between PRIT and SDIF depending on the distance parameter ϵ_Q of the phase clustering algorithm Rough-DBSCAN.

A. STATISTICAL PROPERTIES OF CPRIT

In this section, we share the statistical properties of CPRIT by providing the first, and second moments and the variance of the test statistic. We start with the assumption that the noise in the complex exponentials is independent and identically distributed (i.i.d.) complex Gaussian with $\mathcal{CN}(\mathbf{0}, \mathbf{\Sigma})$, where $\mathbf{\Sigma}$ is the covariance matrix with σ^2 on the diagonal elements and zero otherwise. Let there be L phase clusters of complex exponentials with $N_1, N_2, \dots, N_L$ numbers of contributing pulse pairs and mean values of $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_L$. Here, the vector $\boldsymbol{\mu}_i \in \mathbb{R}^2$ consists of real and imaginary parts of the complex cluster means. Each cluster generates a sum of exponentials with complex Gaussian distribution $\mathcal{CN}(N_i \boldsymbol{\mu}_i, N_i \mathbf{\Sigma})$ and forms the test statistic by taking the absolute of the exponential sum as indicated by (13), which follows the Rician distribution $|\mathcal{CN}(N_i \boldsymbol{\mu}_i, N_i \mathbf{\Sigma})|$. Then, our weighting scheme that is inversely proportional to the variance of these distributions is applied by using the coefficients $\mathbf{w} = [w_1, w_2, \dots, w_L]^T$, as expressed in (14). The final test statistic is formed by fusing the test statistics coming from the clusters, which is the sum of Rician random variables, i.e.,

$$G = \sum_{i=1}^L R_i, \quad (20)$$

where $R_i = |\mathcal{CN}(w_i N_i \boldsymbol{\mu}_i, w_i^2 N_i \mathbf{\Sigma})|$ and its density function is expressed as

$$p(r_i | A, \theta) = \frac{r_i}{\theta^2} e^{-\frac{r_i^2 + A_i^2}{2\theta_i^2}} I_0\left(\frac{A_i r_i}{\theta_i^2}\right), \quad A_i \geq 0. \quad (21)$$

Here, A_i equals $w_i N_i \|\boldsymbol{\mu}_i\|$, θ_i^2 equals $w_i^2 N_i \sigma^2$, and I_0 is the zero-order modified Bessel function of the first kind. The summation of Rician random variables in (20) happens in several engineering strategies, such as envelope combining for multi-antenna receivers or extending the correlation period in GNSS [51]. Although, a closed-form solution is unavailable, some works can accurately approximate the distribution [52]. Moreover, the expected value of the test statistic for the i -th cluster can be found as

$$\begin{aligned} \mathbb{E}[R_i] &= \int r p(r_i | A_i = w_i N_i \|\boldsymbol{\mu}_i\|, \theta_i^2 = w_i^2 N_i \sigma^2) dr_i, \\ &= \theta_i \sqrt{\frac{\pi}{2}} L_{1/2}\left(\frac{-A_i^2}{2\theta_i^2}\right), \\ &= w_i \sqrt{N_i} \sigma \sqrt{\frac{\pi}{2}} L_{1/2}\left(\frac{-(w_i N_i \|\boldsymbol{\mu}_i\|)^2}{2w_i^2 N_i \sigma^2}\right), \end{aligned} \quad (22)$$

where $L_{1/2}(\cdot)$ is the Laguerro polynomial and its series expansion is calculated as

$$L_{1/2}(-x) = \frac{2\sqrt{x}}{\pi} + \frac{1}{2\sqrt{\pi x^{1/2}}} + \frac{1}{16\sqrt{\pi x^{3/2}}} + \epsilon_L, \quad (23)$$

where ϵ_L denotes the negligible terms. Then, the first moment of the final test statistic is found as

$$\begin{aligned} \mathbb{E}[G] &= \sum_{i=1}^L \mathbb{E}[R_i], \\ &= \sum_{i=1}^L \left[w_i \sqrt{N_i} \sigma \sqrt{\frac{\pi}{2}} L_{1/2}\left(\frac{-(w_i N_i \|\boldsymbol{\mu}_i\|)^2}{2w_i^2 N_i \sigma^2}\right) \right]. \end{aligned} \quad (24)$$

Next, second moment of the test statistic for the i -th cluster is found as follows

$$\begin{aligned} \mathbb{E}[R_i^2] &= 2\theta_i^2 + A_i^2, \\ &= 2w_i^2 N_i \sigma^2 + (w_i N_i \|\boldsymbol{\mu}_i\|)^2. \end{aligned} \quad (25)$$

The variance of the final test statistic is then reached as follows

$$\text{var}(G) = \sum_{i=1}^L \text{var}(R_i), \quad (26)$$

$$= \sum_{i=1}^L [\mathbb{E}[R_i^2] - \mathbb{E}[R_i]^2], \quad (27)$$

$$= \sum_{i=1}^L [2w_i^2 N_i \sigma^2 + (w_i N_i \|\boldsymbol{\mu}_i\|)^2 - \mathbb{E}[R_i]^2]. \quad (28)$$

We notice that both the mean and variance of the test statistic generated by CPRIT depend on A and θ^2 values, which come from the statistical features of Rician distribution. Larger values of $\boldsymbol{\mu}_i, N_i, w_i$, and σ^2 result in higher mean and variance for the CPRIT test statistic. Hence, clusters that are comprised of large numbers of similar pulse pairs contribute to the final test statistic more.

B. PROPOSITIONS

In this section, we compare the value of the test statistics of CPRIT, PRIT, and SDIF. We derive lower and upper bounds on the value for two different CPRIT modes. In the first, we utilize CPRIT without the optimal weighting, which implies that an equal weighting scheme is applied, i.e., $\mathbf{w} = \mathbf{1} \in \mathbb{R}^L$. In the second one, CPRIT applies the optimal weighting scheme. We make multiple propositions that individually show the proposed clustering and weighting techniques' effects.

Proposition 1: The test statistic of CPRIT without optimal weighting is lower-bounded by the test statistic of PRIT.

Proof. By utilizing the triangle rule on the test statistic of CPRIT

$$\begin{aligned} G[k] &= \sum_l \left| \sum_{\substack{i \\ \{\theta_{k,i} \in S_l\}}} E_{k,i} \right|, \\ G[k] &\geq \left| \sum_l \sum_{\substack{i \\ \{\theta_{k,i} \in S_l\}}} E_{k,i} \right| = D[k], \\ G[k] &\geq D[k], \end{aligned} \quad (29)$$

we show that the test statistic of CPRIT without is lower-

bounded by the test statistic of PRIT, which completes the proof.

Proposition 2: The test statistic of CPRIT without optimal weighting is upper-bounded by the test statistic of SDIF.

Proof. The test statistic of SDIF can be written as

$$\begin{aligned} H[k] &= \sum_{\substack{(m,n) \\ \{\tau_k - b_k/2 \leq t_n - t_m < \tau_k + b_k/2\}}} \mathbb{1}(m, n, k), \\ &= \sum_l \sum_{\substack{i \\ \{\theta_{k,i} \in S_l\}}} |E_{k,i}|. \end{aligned} \quad (30)$$

Here $\mathbb{1}(\cdot)$ indicates the indicator function. Then, by utilizing the triangle rule on the norm operation

$$H[k] \geq \sum_l \left| \sum_{\substack{i \\ \{\theta_{k,i} \in S_l\}}} E_{k,i} \right| = G[k], \quad (31)$$

we show that the test statistic of CPRIT without weighting is upper-bounded by the test statistic of SDIF, which completes the proof.

Proposition 3: The test statistic of CPRIT with optimal weighting is upper-bounded by the test statistic of CPRIT without weighting.

Proof. The equivalent statement

$$\sum_l w_{k,l} G'[k, l] \leq \sum_l G'[k, l], \quad (32)$$

is true if and only if $w_{k,i} \leq w_{k,j}$ for all $G'[k, i] \geq G'[k, j]$. Our weighting scheme puts more emphasis on the clusters with less variance, i.e., clusters formed by less pulse pairs. Thus, the weighting coefficient gets smaller and smaller as the number of valid pulse pairs or $G'[k, l]$ increases, which completes the proof.

Proposition 4: As the distance parameter ϵ_Q of the phase clustering algorithm gets closer to zero, CPRIT converges to SDIF.

Proof. CPRIT generates C clusters out of C valid pulse pairs if the distance parameter ϵ_Q of the phase clustering algorithm is zero. Additionally, the optimal weighting becomes the equal weighting scheme since each cluster has only one element having zero variance. Therefore, the test statistic of CPRIT becomes the same as SDIF's, which completes the proof.

Proposition 5: As the distance parameter ϵ_Q of the phase clustering algorithm gets closer to infinity, CPRIT converges to PRIT.

Proof. CPRIT generates only one cluster out of C valid pulse pairs if the distance parameter ϵ_Q of the phase clustering algorithm is infinity. Additionally, the weighting has no impact since there is only one cluster. Therefore, the test statistic of CPRIT becomes the same as PRIT's, which completes the proof.

Overall, the phase clustering leads to higher test statistics. At the same time, optimal weighting causes the variance of the test statistic to be minimized at the cost of decreasing the test statistic. Note that CPRIT having a test statistic

that is upper-bounded by the test statistic of SDIF does not indicate that its false alarm rate is absolutely lower than the false alarm rate of SDIF, as the two methods utilize different threshold functions. However, in our experiments, we observe that CPRIT mostly delivers similar or lower false alarm rates to SDIF. Compared to PRIT, CPRIT is shown to yield higher correct detection rates while not introducing a vital disadvantage.

Furthermore, we infer that CPRIT plays a role between PRIT and SDIF. As the distance parameter ϵ_Q of the Rough-DBSCAN gets closer to zero, (14) would produce the same test statistic value as the SDIF's. Similarly, as ϵ_Q gets closer to infinity, (14) would produce the same test statistic value as the PRIT's.

C. COMPLEXITY ANALYSIS

Let N be the number of pulses. Rough-DBSCAN works in two parts: course clustering and fine matching. The first part is always utilized with the distance parameter d in the $[0, \epsilon_Q]$ range, in which ϵ_Q denotes the distance parameter of the second part. Depending on whether d is close to the minimum or maximum of the range, the complexity of the clustering operation becomes $\mathcal{O}(N^2)$ or $\mathcal{O}(N)$, respectively. SDIF has the complexity that is between $\mathcal{O}(N)$ and $\mathcal{O}(N^2)$, depending on the maximum number of differences k_{max} allowed. As k_{max} increases, the complexity rises. PRIT-I and PRIT-II have roughly the complexity of $\mathcal{O}(N^2)$, whereas PRIT-II has slightly more computations due to the offset correction. Traditional methods employ a two-stage approach, including the RF-PW-AOA clustering and TOA analysis with SDIF or PRIT. Therefore, considering both stages, they have $\mathcal{O}(N^2)$ complexity. CPRIT utilizes Rough-DBSCAN in the phase clustering phase and employs PRIT in the TOA analysis part. Since there are $N(N-1)/2$ pulse pairs to be combined, it has a complexity between $\mathcal{O}(N^2)$ and $\mathcal{O}(N^4)$, depending on the value of d . Overall, the proposed method exhibits more complexity, as it aims to combine all pulse pairs with distance weighting.

V. NUMERICAL RESULTS

The numerical analyzes on simulation data consists of two parts: the test statistics value analysis and the detection performance analysis. The first part demonstrates the increase achieved in the test statistic of PRI transform when the proposed method is deployed. The second part examines the detection performances of SDIF, PRIT, and CPRIT for different PRI types. In our analyses, we work on synthetically generated pulses to be detected via the receivers operating on the amplitude of the downconverted signal. Note that having a detection algorithm working on the amplitude of the pulses provides sufficient resistance to the majority of pulse modulation techniques (MoP), i.e., frequency and phase modulation [2], [53], [54]. Since channelized receivers are used in the most ESM applications, pulse-on-pulse situation is not a problem as long as they are detected by different channels. In other words, pulses with different frequencies

can be deinterleaved, even though they are intercepted at the same time. In this section, we work on already detected pulses with no MoP and we do not assume PoP situation. Therefore, we will not discuss the details of the pulse detection process, as our primary focus is the deinterleaving process only. We also ignore the propagation and system effects, such as phase shifts occurring in the signal samples, as their effects are mostly encountered before the deinterleaving procedure. More details regarding the simulation setup are shared in the following subsections.

A. TEST STATISTICS VALUE ANALYSIS

This analysis aims to demonstrate the increase in the test statistics values when the proposed approach is deployed. By doing this, we also show why CPRIT provides better detection performance. The results are taken under the Gaussian noise with zero mean and 10 nanoseconds standard deviation that is independently and identically added to the TOA values. We assume that all radars have the same RF-PW-AOA values, meaning that $\psi_{k,j} = 1$ for all k, j and there is no missing pulse since we would like to put forward the phase changes and their impacts more clearly. We investigate the PRI types indicated in Figure 1 to demonstrate the phase toggle and its effect on the test statistics. The proposed approach is applied on PRIT-I and PRIT-II, which we denote as CPRIT-I and CPRIT-II, respectively. The test statistics of the methods are analyzed via the output-input ratio (OIR). It is calculated as

$$OIR_k = \frac{D[k]}{C_k}, \quad (33)$$

for PRIT-I and PRIT-II,

$$OIR_k = \frac{G[k]}{C_k}, \quad (34)$$

for the proposed method. Here, C_k is the total number of valid pulse pairs, with the TOA values (t_n, t_m) contributing to the test statistic of k -th PRI level. The metric OIR is particularly chosen to demonstrate the destructive effect observed when the toggled exponentials are summed in PRIT. It indicates the ratio of pulses a PRI estimator can use to generate test statistics. Hence it ranges from zero to one. It is desired to have an OIR value that is close to one, which means that the phases of the exponentials do not vary much and create a constructive impact when they are summed. Hence, the test statistic will most likely stay above the threshold, and missed detections will not happen. Since the constant PRI type has the least variation, all phase angles match, and OIR gets closer to one. However, if PRI values vary, as in the case of staggered and jittered types, the phase angles do not match, and OIR reduces.

As time passes, the receiver intercepts more pulses from radars. Hence, the number of valid pulse pairs for the PRI levels corresponding to the existing radars increases in time. The more the number of valid pulse pairs is, the more the accumulative phase noise is. In the next subsections, we analyze the variation of the phase angles in the PRI transform

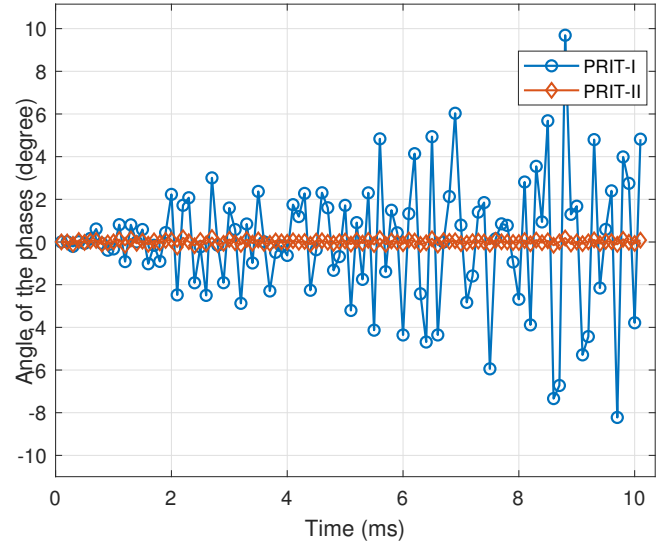


Figure 7. Angle of the phases (degree) in the case of constant PRI.

methods for different PRI types as the number of valid pulse pairs increases. In this way, we observe the impact of the accumulative phase noise in the test statistic. Additionally, we compare the proposed approach CPRIT with SDIF and PRIT according to their OIR results.

1) Constant PRI Case

Both PRI transform methods (PRIT-I, PRIT-II) are known to perform well in the case of constant PRI. Since the angles of the complex exponentials match each other and become constructive, the test statistic is most likely to stay above the threshold. Figure 7 demonstrates the angle of the exponentials corresponding to the radar with 100 μs constant PRI previously shown by Figure 1. It is observed that PRIT-I starts to have variations in the angle values after only a long measurement duration due to the accumulation of noise in TOAs. Since PRIT-II was specifically developed to deal with this drawback occurring much more in the jittered PRI case, it also corrects the phase jumps in the constant PRI case.

Figure 8 shows the OIR change as the number of valid pulse pairs varies. Since both PRIT-I and PRIT-II perform well in the constant PRI case, the proposed approaches CPRIT-I and CPRIT-II only advance their results a little. Even though PRIT-I has a slight decrease in OIR as the number of pairs increases, it is observed that all of the methods always have $OIR \approx 1$.

2) Staggered PRI Case

PRI transform is known to be not suited for the staggered PRI since the angles of the exponentials constantly toggle and have a destructive effect on the test statistic, which yields missed detections. In Figure 9, we demonstrate the phase toggles corresponding to the radar with [100, 140, 170] μs staggered PRI previously shown in Figure 1. The PRI sequence has three levels; therefore, there are three clusters of

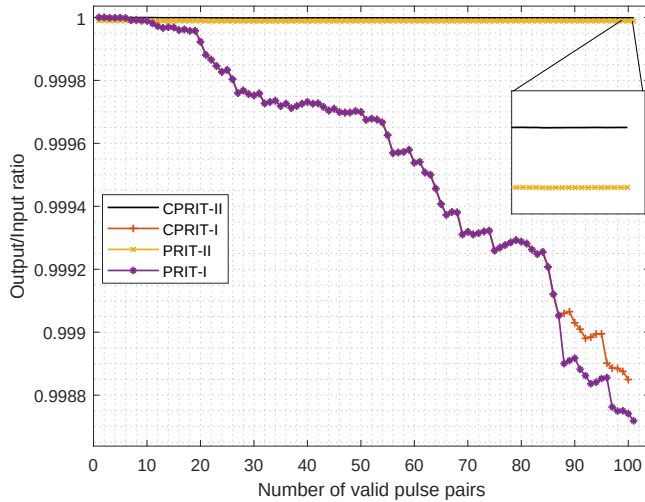


Figure 8. OIR vs number of valid pulse pairs in the case of the constant PRI.

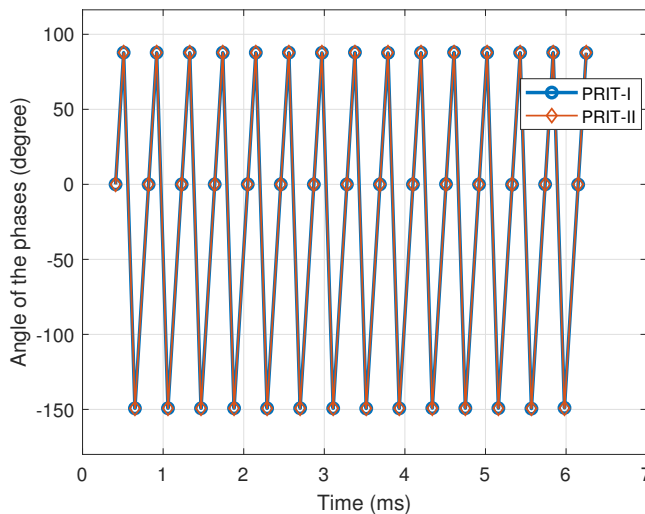


Figure 9. Angle of the phases (degree) in the case of staggered PRI.

the phases, if the PRI transform is calculated at the sequence period $410 \mu s$ (τ_k in (5) is set to $410 \mu s$). Since the phase jumps are pretty high, the destructive effect on the test statistic is high, and PRIT-II cannot compensate for this drawback, as shown in Figure 10. Therefore, the test statistics of both PRI transform methods get drastically lower as the number of valid pulse pairs increases, which causes the missed detection. The proposed method removes this destructive effect by working with matching phases coming from similar exponentials in each cluster. Hence, both CPRIT-I and CPRIT-II provide $OIR \approx 1$ without depending on the number of valid pulse pairs.

3) Pulse Group Constant (PGC) PRI Case

PGC PRI type has a similar pattern to the staggered PRI since both have a periodic sequence. However, unlike the staggered PRI, PGC includes bursts of pulses and long silences between

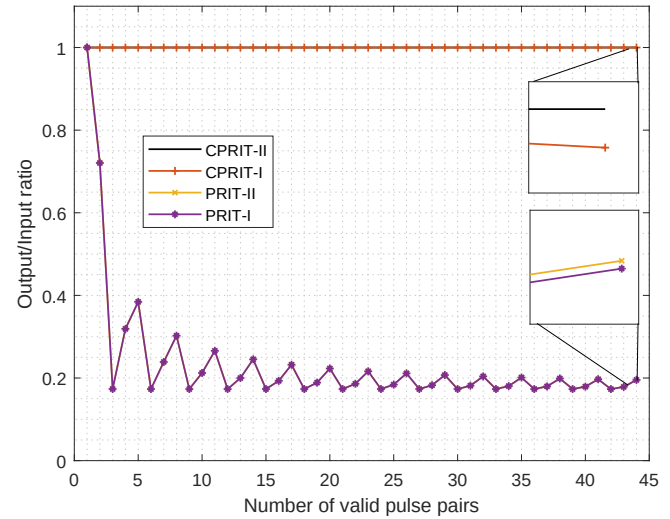


Figure 10. OIR vs number of valid pulse pairs in the case of the staggered PRI.

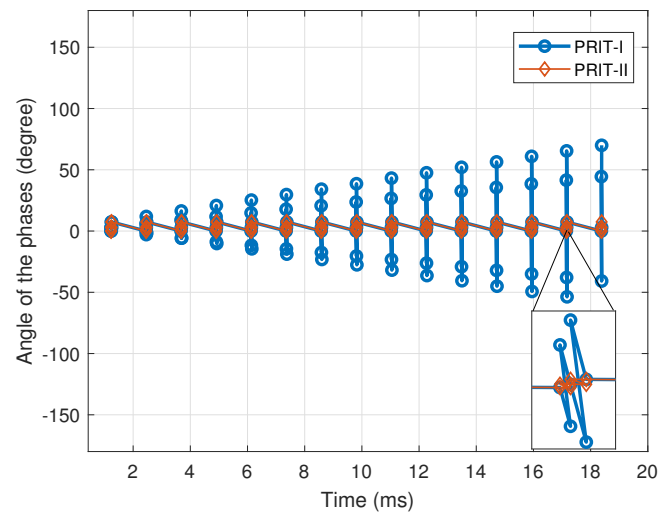


Figure 11. Angle of the phases (degree) in the case of PGC PRI.

the bursts. Therefore, there are very low and high levels in the PRI sequence. In Figure 11, we demonstrate the phase toggles corresponding to the radar with $[10, 15, 1200] \mu s$ PGC PRI previously shown in Figure 1. The PRI transform is calculated at the sequence period $1225 \mu s$ (τ_k in (5) is set to $1225 \mu s$). Low PRI values in the sequence cause insignificant phase changes that do not cause a destructive effect since variations on the term $t_n/(t_n - t_m)$ in (5) are quite low. However, phase changes get more prominent as the number of valid pulse pairs increases, or more measurements are taken. This issue can also be verified by the drop of PRIT-I test statistic in Figure 12, while PRIT-II can compensate for small phase jumps and provides $OIR \approx 1$. CPRIT-I also removes the effect of phase jumps occurring in PRIT-I, and it provides $OIR \approx 1$ without depending on the number of valid pulse pairs.

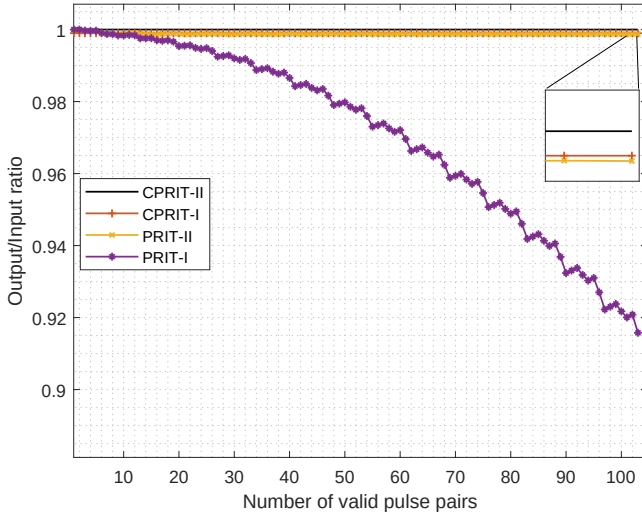


Figure 12. OIR vs number of valid pulse pairs in the case of the PGC PRI.

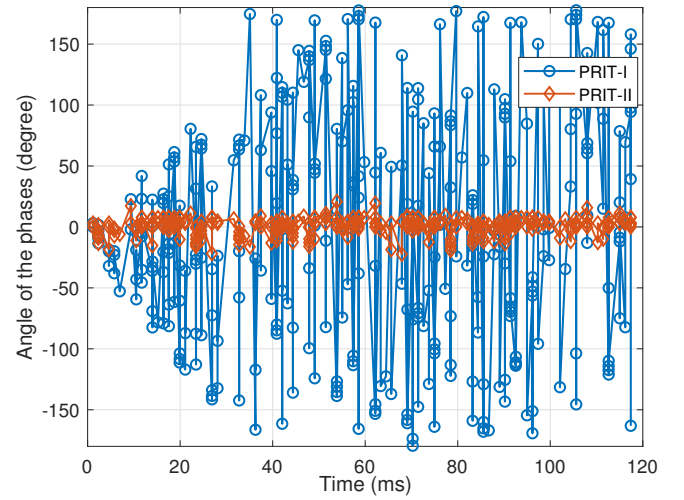


Figure 13. Angle of the phases (degree) in the case of PGJ PRI.

4) Pulse Group Jittered (PGJ) PRI Case

PGJ PRI type is similar to PGC PRI, as both transmit the pulses in bursts. However, unlike the PGC, PGJ has a varying PRI value in its pattern, which is usually the largest PRI value indicating the silences between the bursts. In Figure 13, we demonstrate the phase toggles corresponding to the radar with $[10, 15, x]$ μs PGJ PRI previously shown in Figure 1. Here the largest PRI value x is distributed uniformly in the range of $[1100, 1200]$ μs . PGJ PRI is different from constant, staggered, and PGC PRI types in a way that its sequence is random and not periodic. Therefore, the PRI transform is not calculated at the sequence period. Instead, it is utilized on the mean value of the PRI sequence 1175 μs (τ_k in (5) is set to 1175 μs). Since one of the PRI values in the sequence is varying, similar to the jittered PRI case, there is a significant deviation in the phase angle of the exponentials, which is a problem for PRIT-I. In Figure 14, it is shown that PRIT-I yields a very low test statistic as the number of valid pulse pairs increases due to the destructive effect of summing exponentials with distinct phases. The proposed approach CPRIT-I organizes these exponentials, leading to a much larger test statistic with $OIR \approx 1$. PRIT-II can also compensate for phase transitions, and it provides $OIR \approx 1$ as well. Hence, only PRIT-I fails to generate large enough test statistics to enable the detection of PGJ.

5) Jittered PRI Case

In the case of jittered PRI, phase jumps are smaller than the staggered PRI and more prominent than the constant/PGC PRI cases. In Figure 15, we demonstrate the phase toggles corresponding to the radar with jittered PRI. The PRI of the radar is uniformly distributed in the range $[90, 110]$ μs previously shown in Figure 1. Similar to PGJ PRI, PRI transform is utilized on the mean value of the PRI sequence. Compared to PRIT-I, PRIT-II results in phases with much more minor

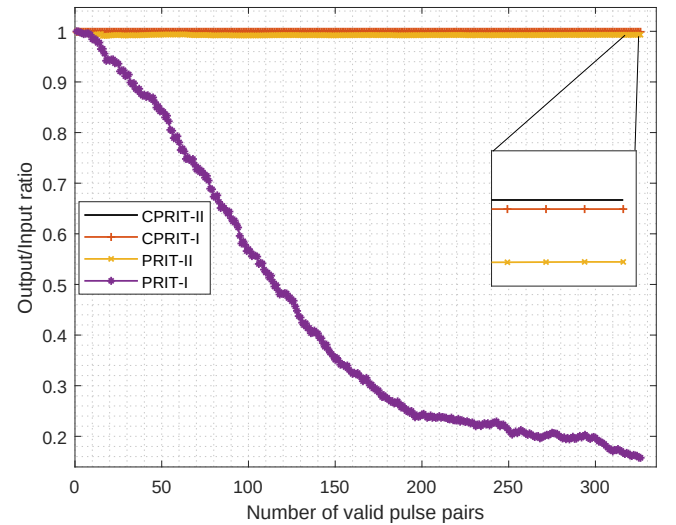


Figure 14. OIR vs number of valid pulse pairs in the case of the PGJ PRI.

deviations; however, it partially removes the phase jumps, and there are still a considerable amount of fluctuations. Figure 16 indicates that PRIT-II succeeds in providing a higher test statistic compared to PRIT-I. Both CPRIT-I and CPRIT-II improve the test statistics of the PRI transform methods and provide $OIR \approx 1$ without depending on the number of valid pulse pairs.

B. DETECTION PERFORMANCE ANALYSIS

This section demonstrates the detection performances of CPRIT, PRIT, and SDIF. Four different missing pulse percentages $\in \{0, 10, 20, 30\}$ and seven different radar scenarios, including different PRI-RF-PW-AOA types, are considered. 1000 Monte Carlo simulations are conducted to obtain the numerical results. Gaussian noise with zero mean and 10 nanoseconds standard deviation is independently and identi-

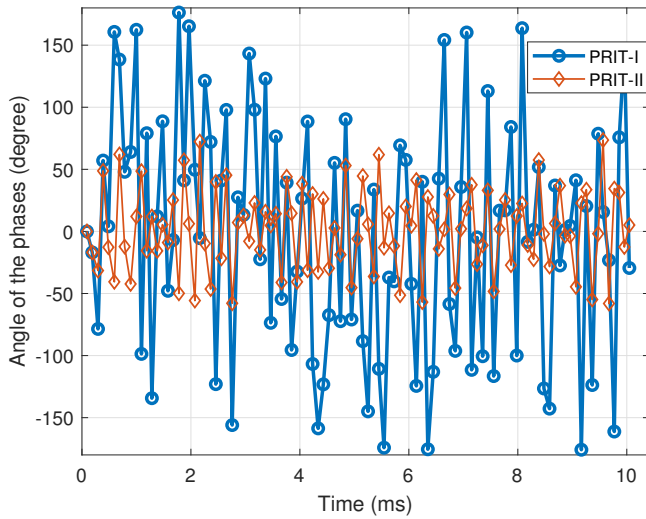


Figure 15. Angle of the phases (degree) in the case of jittered PRI.

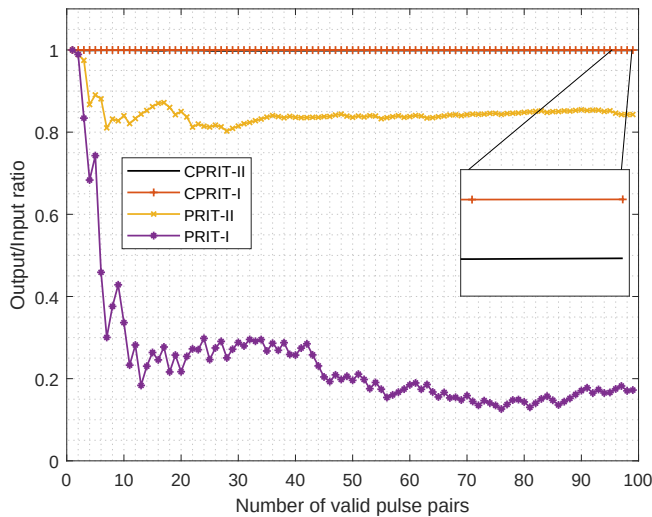


Figure 16. OIR vs number of valid pulse pairs in the case of jittered PRI.

cally added to the TOA values in each run.

In the first five scenarios, we aim to analyze the TOA analysis performances of the methods according to the PRI types. Hence, radars with the same PRI types but different values simultaneously exist. Their RF-PW-AOA types and values are also the same so that the clustering stage has no impact on the TOA analysis part. The sixth scenario is a mixed PRI case, while RF-PW-AOA parameters of the radars remain the same. PRI types and values vary, therefore, this scenario shows the methods' overall TOA analysis performance. The last scenario is a fully mixed case, in which radars with different PRI-RF-PW-AOA parameter types and values exist. It reveals both clustering and TOA analysis performance of the traditional and proposed methods and it resembles a real EW environment. Throughout this analysis, the number of pulses N per radar varies $\in [7, 50]$, so the deinterleaving

problem has enough complexity to generalize our results.

Regarding detection performances, we compare CPRIT-I, CPRIT-II, PRIT-I, PRIT-II, and SDIF. As these methods are PRI estimators, they must be followed by a sequence retrieval algorithm to separate the pulses and obtain their performance results. In our experiments, all methods are followed by a moving average sequence retrieval algorithm, which solves the cumulative error resulting from PRI jitter [28]. It updates the searched PRI value each time a pulse is separated until no pulse is left.

Correct detection and false alarm rates are used as the metrics. Mathematically, the correct detection rate can be calculated as

$$\frac{\text{True positive}}{\text{True positive} + \text{False negative}}, \quad (35)$$

and the false alarm rate can be found as

$$\frac{\text{False positive}}{\text{False positive} + \text{True negative}}. \quad (36)$$

A pair of pulses from the same emitter is considered as a true positive if the deinterleaving algorithm outputs them in the same group. If they are in different groups, it is considered as a false negative. Similarly, a pair of pulses from different emitters is considered as a true negative if the deinterleaving algorithm outputs them in different groups. If they are in the same group, it is considered as a false positive.

1) Scenario-I: Constant PRI Case

The next five scenarios including this one aim to analyze the deinterleaving performance of the methods depending on the PRI type. Thus, we keep the RF-PW-AOA parameters of the radars the same, while they have the same PRI type with different values. In this scenario, three radars exist simultaneously with constant PRI in the range of $[100, 200] \mu\text{s}$. In each Monte Carlo run, center PRI values of the radars are drawn from the given range, and synthetic data is created after the TOAs of the radars are interleaved in the time axis.

Figure 17, and 18 demonstrate the methods' correct detection and false alarm performances against the missing pulse percentage, respectively. Since all radars have constant PRI, the PRI spectrum is occupied less. The methods have similar detection performance when the missing pulse percentage is low. However, the detection performance of SDIF deteriorates as the missing pulse percentage increases. Furthermore, it is observed that the methods yield low false alarm rates without depending on the missing pulse percentage. Note that false alarm rates are inversely proportional to the missing pulse percentage, as the methods provide lower test statistics that stay below the threshold due to missing pulses. Thus, both correct detection and false alarm rates usually decrease if the missing pulse percentage increases.

2) Scenario-II: Staggered PRI Case

In this scenario, three radars exist simultaneously with staggered PRI sequences having lengths $\in [2, 16]$ and values $\in [100, 200] \mu\text{s}$. In each Monte Carlo run, PRI values and the

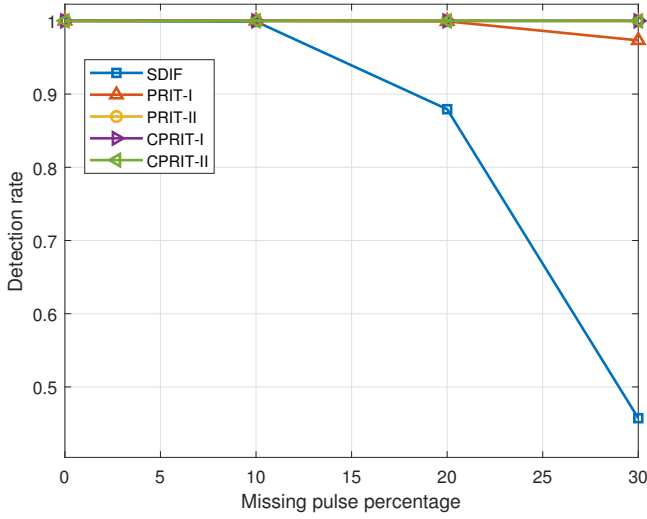


Figure 17. Correct detection rate in the case of constant PRI.

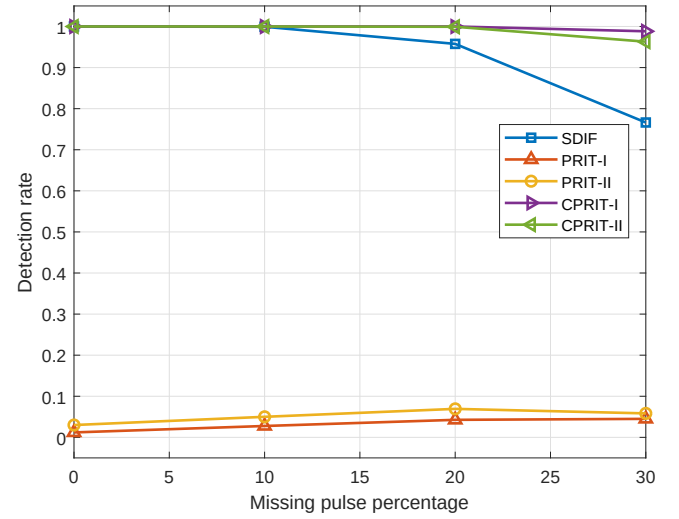


Figure 19. Correct detection rate in the case of staggered PRI.

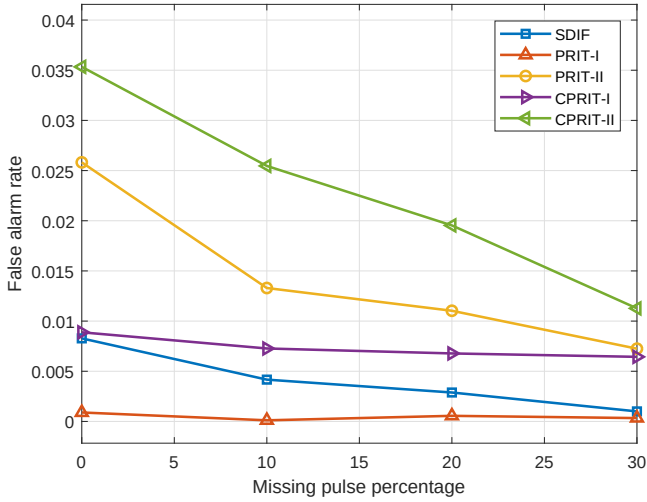


Figure 18. False alarm rate in the case of constant PRI.

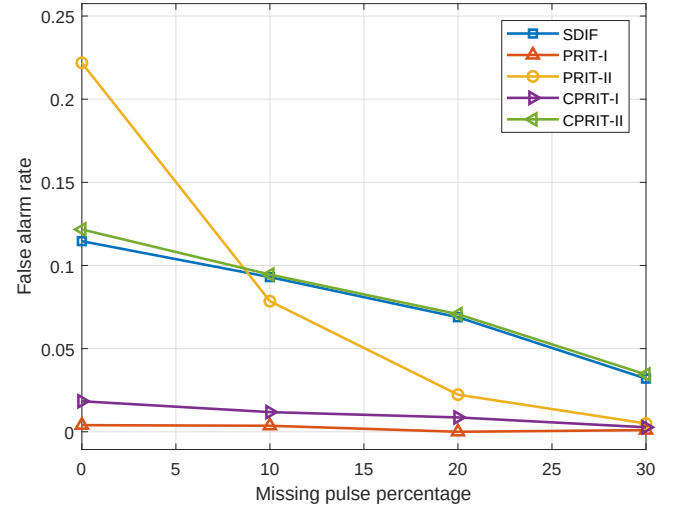


Figure 20. False alarm rate in the case of staggered PRI.

sequence length of the radars are drawn from the given ranges, and synthetic data is created after the TOAs of the radars are interleaved in the time axis. Figure 19, and 20 demonstrate the methods' correct detection and false alarm performances against the missing pulse percentage, respectively. Since the complex autocorrelation functions of PRIT-I and PRIT-II are comprised of exponentials with toggling phases, their test statistics become low and stay below the threshold, which results in a missed detection. Furthermore, CPRIT-I and CPRIT-II enable PRI transform to have acceptable detection rates and perform better than SDIF at the cost of slightly higher false alarm rates.

3) Scenario-III: PGC PRI Case

In this scenario, three radars exist simultaneously with PGC PRI sequences having lengths $\in [2, 5]$. PRI values are chosen from the intervals $[5, 50]$ μs inside the burst and $[1000, 10000]$

μs between the bursts. In each Monte Carlo run, PRI values and the sequence length of the radars are drawn from the given ranges, and synthetic data is created after the TOAs of the radars are interleaved in the time axis. Figure 21, and 22 demonstrate the methods' correct detection and false alarm performances against the missing pulse percentage, respectively. Unlike the staggered PRI case, the test statistics provided by PRIT-I and PRIT-II are sufficiently large. Therefore, all methods achieve the detection of the PGC PRI. SDIF yields the lowest detection rate, as its performance drastically reduces at high missing pulse percentages. It is also observed that both CPRIT-I and CPRIT-II provide lower false alarm rates than PRIT-I and PRIT-II.

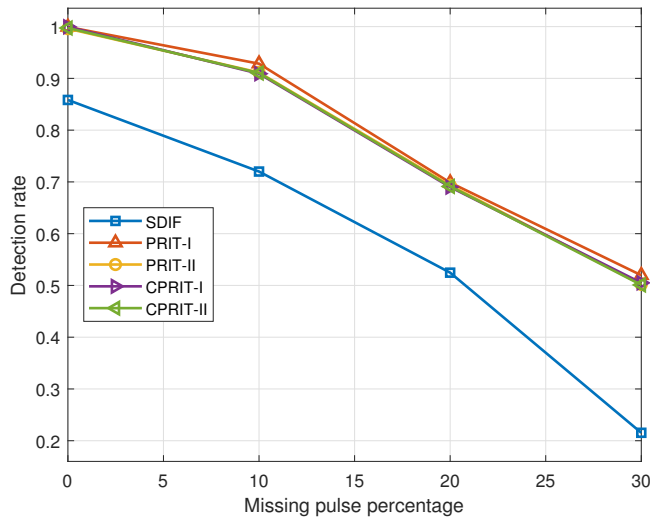


Figure 21. Correct detection rate in the case of PGC PRI.

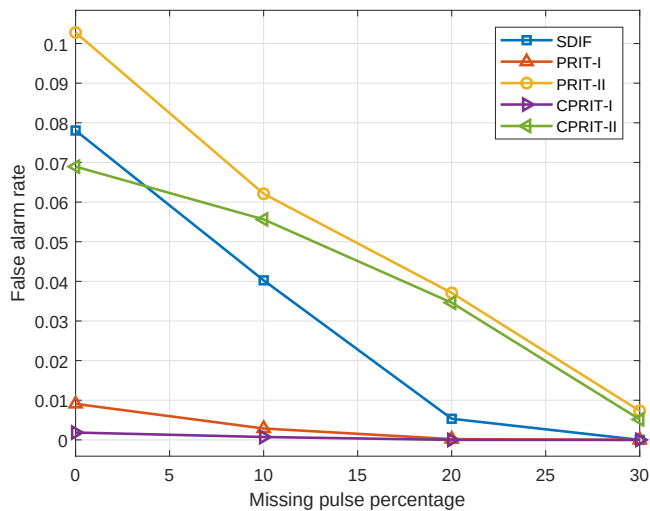


Figure 22. False alarm rate in the case of PGC PRI.

4) Scenario-IV: PGJ PRI Case

In this scenario, three radars exist simultaneously with PGJ PRI sequences having lengths $\in [2, 5]$. Inside the burst, the PRI values are chosen from the interval $[5, 50] \mu s$. Denote the PRI values between two sequential bursts as X . It is drawn from the interval $[1000, 10000] \mu s$, then uniformly distributed variation in the range of $[-0.1X, 0.1X]$ is randomly added after each repetition of the sequence. In each Monte Carlo run, PRI values and the sequence length of the radars are drawn from the given ranges, and synthetic data is created after the TOAs of the radars are interleaved in the time axis. Figure 23, and 24 demonstrate the methods' correct detection and false alarm performances against the missing pulse percentage, respectively. Different from the PGC PRI case, having fluctuating complex exponentials causes PRIT-I to have the lowest correct detection rates. CPRIT-I increases the correct

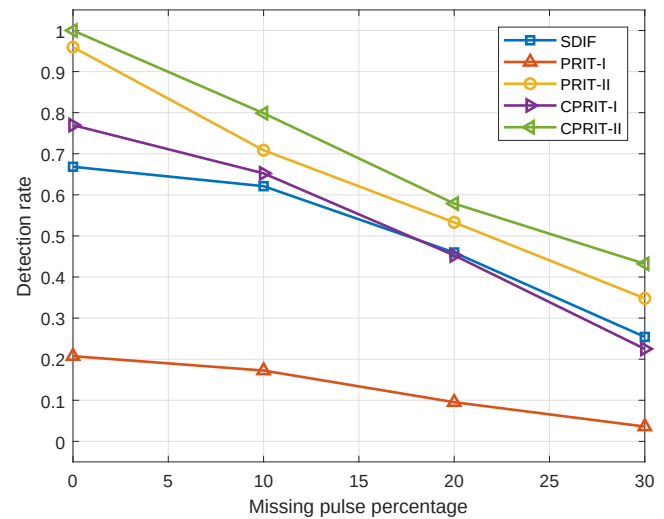


Figure 23. Correct detection rate in the case of PGJ PRI.

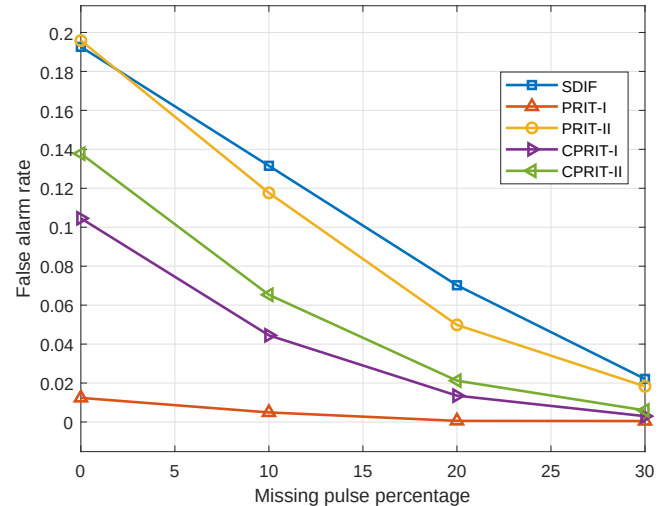


Figure 24. False alarm rate in the case of PGJ PRI.

detection rate of PRIT-I to a certain level. However, it is outperformed by PRIT-II. CPRIT-II performs the best, as it works on exponentials whose phase jumps are compensated more. We also observe that CPRIT-I has higher false alarm rates than PRIT-I, while CPRIT-II provides lower false alarm rates than PRIT-II. Lastly, SDIF achieves mediocre detection and false alarm results.

5) Scenario-V: Jittered PRI Case

In this scenario, three radars exist simultaneously with jittered PRI, whose center level $\in [100, 200] \mu s$ and PRI jitter ratio $\in [0.1, 0.3]$. In each Monte Carlo run, center PRI values and the jitter ratios of the radars are drawn from the given ranges, and synthetic data is created after the TOAs of the radars are interleaved in the time axis. Note that the jittering effect is realized as a uniform distribution that is added to the center

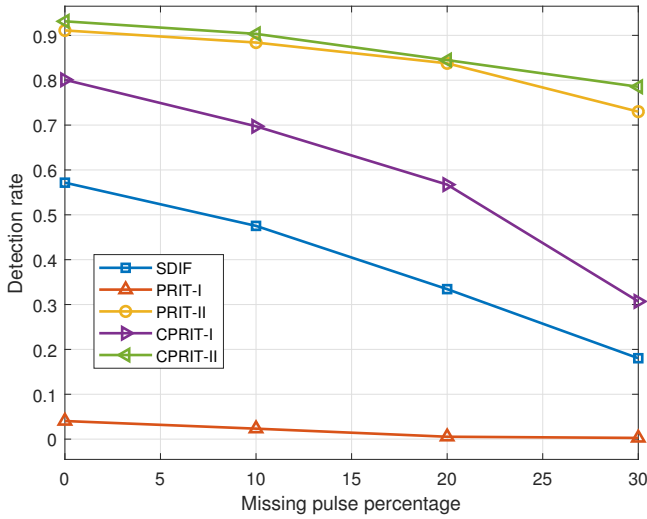


Figure 25. Correct detection rate in the case of jittered PRI.

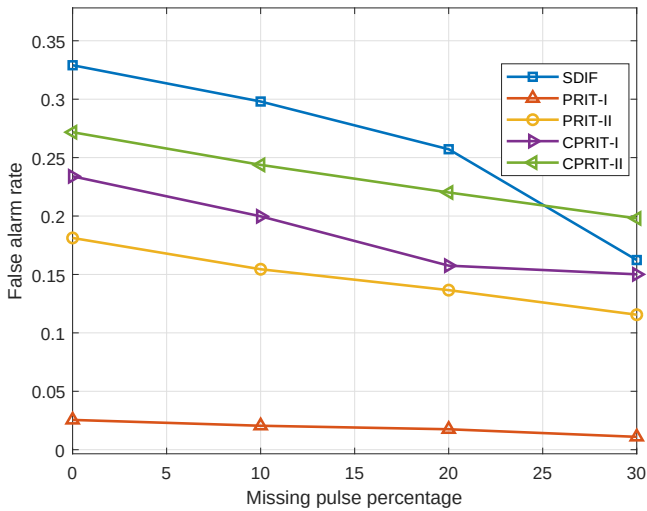


Figure 26. False alarm rate in the case of jittered PRI.

PRI level. Figure 25, and 26 demonstrate the methods' correct detection and false alarm performances against the missing pulse percentage, respectively. Since PRIT-I cannot track sharp phase jumps and cannot produce complex exponentials with similar phases, its test statistic becomes low and causes missed detection. However, CPRIT-I provides the detection as it organizes the exponentials and generates a much higher test statistic. PRIT-II can detect the jittered PRI type, but it is outperformed by CPRIT-II, which has slightly higher false alarm rates. Lastly, SDIF achieves mediocre detection and false alarm results.

6) Scenario-VI: Mixed PRI Case

In this scenario, three radars exist simultaneously with random PRI types described in Section II-A, while the RF-PW-AOA parameters of the radars are the same. Hence, we focus

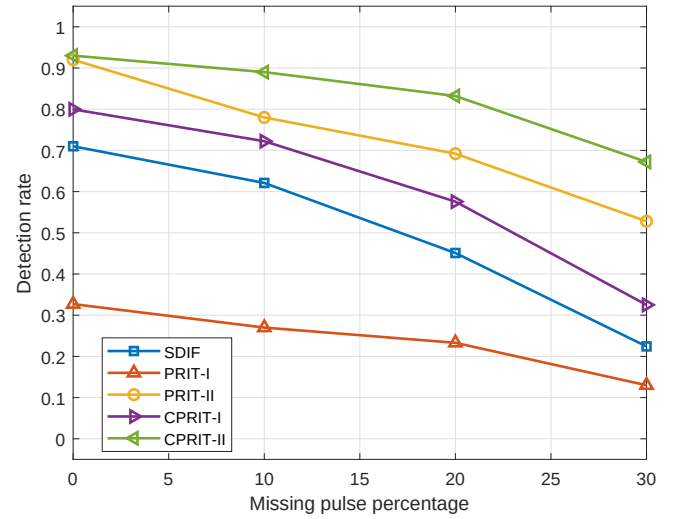


Figure 27. Correct detection rate for the mixed PRI case.

on the overall TOA analysis performance of the methods without caring about the clustering impact. PRI types of radars are randomly picked with 1/5 probability, as there are five types. PRI values depending on the type are drawn as described in the previous sections. In each Monte Carlo run, the TOAs of the radars are interleaved in the time axis.

Figure 27, and 28 demonstrate the methods' correct detection and false alarm performances against the missing pulse percentage, respectively. The proposed approach enables PRIT-I and PRIT-II to reach higher correct detection rates, while SDIF has a mediocre detection result. It is interesting to note that CPRIT-I has higher false alarm rates than PRIT-I, contrary to CPRIT-II which has lower rates than PRIT-II. This is because PRIT-I cannot track phase jumps occurring in varying PRI patterns, such as jittered and PGJ, so it usually provides the lowest test statistic. Therefore, its false alarm rate is the lowest. However, even a slight increase in the test statistic might result in higher false alarm rates, which we observe in CPRIT-I. Moreover, CPRIT-II does not change the test statistic value of PRIT-II since PRIT-II can track phase jumps and generate high enough test statistics except for the staggered PRI case. Hence, CPRIT-II and PRIT-II have similar false results that favor the former. Furthermore, SDIF has the highest false alarm as its performance degrades due to varying PRI patterns and high missing pulse percentages.

7) Scenario-VII: Mixed PRI-RF-PW-AOA Case

This case is a fully mixed case resembling a real EW environment, in which all PRI-RF-PW-AOA parameters (types, values) of the radars differ. Hence, we test the overall deinterleaving performance of the methods, which is affected by both the clustering and TOA analysis stages. The clustering algorithm for the traditional deinterleaving approaches using PRIT and SDIF methods is chosen as the Rough-DBSCAN [47], which we described in Section III-C. The algorithm first clusters AOA, then RF, and lastly PW parameters of the

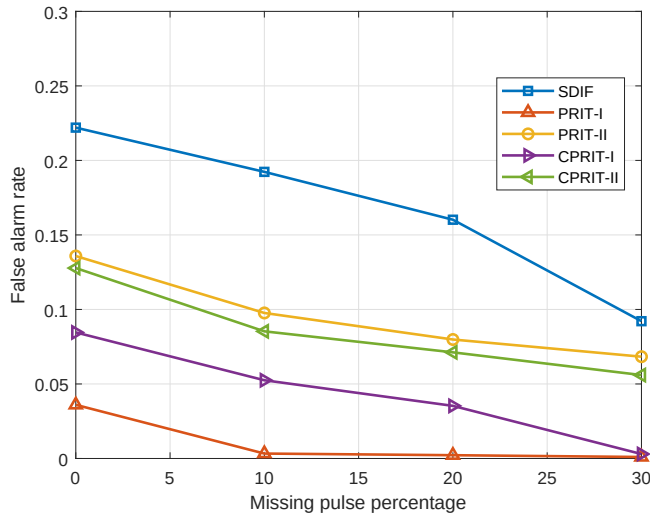


Figure 28. False alarm rate for the mixed PRI case.

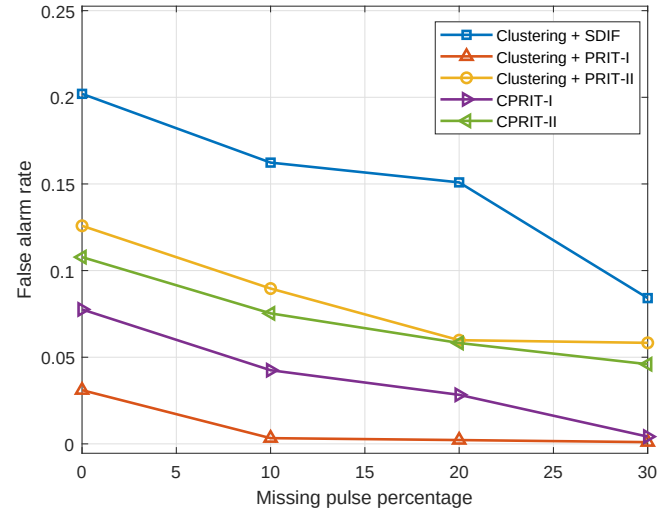


Figure 30. False alarm rate for the fully mixed case.

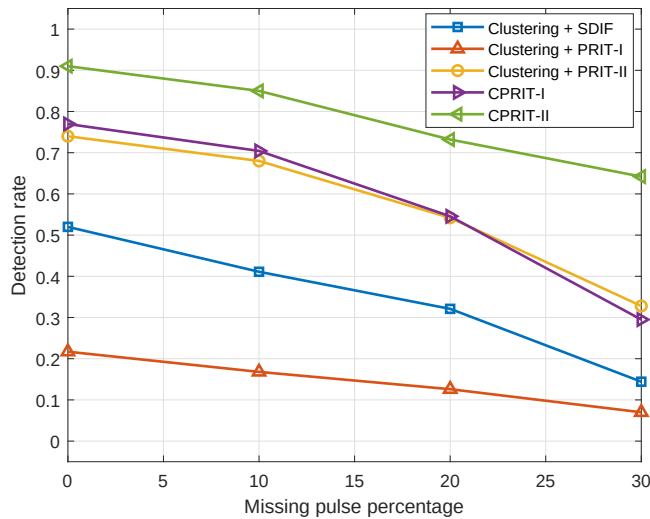


Figure 29. Correct detection rate for the fully mixed case.

pulses. The same distance coefficients in (10) are used with Euclidean distance, and the minimum number of samples is set as three.

Three radars exist simultaneously with random PRI-RF-PW-AOA types described in the previous sections. RF, PW, and AOA values are chosen with uniform distribution from the interval $[2, 18]$ GHz, $[1, \min(100, (PRI))] \mu s$, $[0, 360]$ degrees, respectively. Zero mean Gaussian noise with the standard deviations $\sigma_{RF} = 5$ MHz, $\sigma_{PW} = 0.01 \mu s$, and $\sigma_{AOA} = 5$ degrees are independently and identically added to the RF, PW, AOA parameters of the PDWs, respectively. In the simulations, PRI types of radars are randomly picked with $1/5$ probability, as there are five types, while RF-PW types are randomly picked with $1/3$ probability, as there are three types. In each Monte Carlo run, the TOAs of the radars are interleaved in the time axis.

Figure 29, and 30 demonstrate the methods' correct detection and false alarm performances against the missing pulse percentage, respectively. Compared to Figure 27, on one hand, we observe that the correct detection rates of the traditional approaches utilizing RF-PW-AOA clustering get significantly worse. This is expected, as the clustering algorithm produces incorrect results by over-separating the pulses in the case of high noise and the hopping RF-PW type. On the other hand, we notice that there is only a slight decrease in the detection rates of CPRIT-I and CPRIT-II, thanks to the usage of distance weighting instead of clustering. The distance weighting in (10) combines all pulse pairs, even though they are very distant in RF-PW-AOA parameter space. Hence, larger test statistics values can be reached in the case of high noise and the hopping RF-PW type, which enables the detection. Moreover, Figure 30 shows that all methods' false alarm rates drop similarly when it is compared to Figure 28. Since pulses have different RF-PW-AOA values, all methods tend to develop smaller test statistic values that are below the threshold. This results in less numbers of hypothesis generation before the pulse tracking stage. Therefore, both false alarm and detection rates become lower.

In summary, traditional methods suffer from the two-stage approach including RF-PW-AOA clustering and TOA analysis. Due to noise and hopping RF-PW type, the clustering stage produces erroneous results, which undermines the performance of TOA analysis. Additionally, phase toggling in the staggered PRI is problematic in these methods, which results in missed detection. However, the proposed approach combines all pulse pairs with distance weighting and enables the detection of staggered PRI by clustering the angles of the phases in complex autocorrelation. The overall detection ability of the methods depending on the PRI type is summarized in Table 1.

Table 1. Detection Ability of the Methods [X: poor, O: average, ✓: excellent] for Different PRI Types

Methods Types	PRIT-I	PRIT-II	SDIF	CPRIT-I	CPRIT-II
Constant	✓	✓	O	✓	✓
Staggered	X	X	✓	✓	✓
PGC	✓	✓	O	✓	✓
PGJ	X	O	O	✓	✓
Jittered	X	O	O	✓	✓

VI. CONCLUSIONS

In this paper, we propose Combined PRI Transform (CPRIT) to alleviate the suboptimality in the design of two-stage traditional deinterleaving algorithms and remove the problem of missed detections that occur when PRI transform (PRIT) is used against the staggered PRI type. The proposed method does not utilize RF-PW-AOA clustering, instead applies an inverse exponential distance weighting to combine all pulse pairs, which is necessary against radars with hopping RF-PW parameters. Then, it clusters the phase angles of the complex exponentials in the autocorrelation function. Each cluster calculates its test statistic value independently so that the destructive effect of summing exponentials with distinct phases does not occur. To fuse the test statistics from clusters, we apply an optimal weighting function that minimizes the variance of the estimator. Hence, the final test statistic is reached and compared to the threshold. We theoretically explain how the proposed phase clustering and optimal weighting schemes enable CPRIT to have better detection performance while not introducing a critical disadvantage. It is shown that the phase clustering leads to a higher test statistic value, while the optimal weighting scheme causes the variance of the test statistic to be minimized. Additionally, we mathematically show that CPRIT plays a role between PRIT and SDIF depending on the distance parameter of the phase clustering algorithm.

The numerical experiments compare CPRIT to SDIF and the two versions of PRI transform via Monte Carlo analysis. Seven scenarios including radars with different PRI-RF-PW-AOA parameters are considered. We observe that CPRIT almost always provides the highest correct detection rates, and it enables the detection of the staggered PRI type.

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